Host and Port Scanning

Once we undertand that the Host is alive, we will need more information from that point:

- Open ports and its services
- Service versions
- Information that the services provided
- Operating system

When we scan a port we can find it in 6 different possible states:

`open`	This indicates that the connection to the scanned port has been established. These connections can be **TCP connections**, **UDP datagrams** as well as **SCTP associations**.
`closed`	When the port is shown as closed, the TCP protocol indicates that the packet we received back contains an `RST` flag. This scanning method can also be used to determine if our target is alive or not.
`filtered`	Nmap cannot correctly identify whether the scanned port is open or closed because either no response is returned from the target for the port or we get an error code from the target.
`unfiltered`	This state of a port only occurs during the **TCP-ACK** scan and means that the port is accessible, but it cannot be determined whether it is open or closed.
`open|filtered`	If we do not get a response for a specific port, `Nmap` will set it to that state. This indicates that a firewall or packet filter may protect the port.
`closed|filtered`	This state only occurs in the **IP ID idle** scans and indicates that it was impossible to determine if the scanned port is closed or filtered by a firewall.

Discovering Open TCP Ports

By default, NMAP scans the top 1000 TCP ports with the SYN scan (-sS). This SYN scan is set only to default when we run it as root because of the socket permmissions required to create raw TCP packets. Otherwise, the TCP scan (-sT) is performed by default. This means that if we do not define ports and scanning methods, these parameters are set automatically. We can define the

ports ony by one (-p 22,25,80,139,445), by range (-p 22-445), by top ports (—top-ports-10) from the Nmap database that have been signed as most frequent, by scanning all ports (-p-) but also by defining a fast port scan, which contains top 100 ports (-F).

Scanning Top 10 TCP Ports

```
pecna@htb[/htb]$ sudo nmap 10.129.2.28 --top-ports=10

Starting Nmap 7.80 ( https://nmap.org ) at 2020-06-15 15:36
CEST
Nmap scan report for 10.129.2.28
Host is up (0.021s latency).

PORT     STATE    SERVICE
21/tcp   closed   ftp
22/tcp   open     ssh
23/tcp   closed   telnet
25/tcp   open     smtp
80/tcp   open     http
110/tcp  open     pop3
139/tcp  filtered netbios-ssn
443/tcp  closed   https
445/tcp  filtered microsoft-ds
3389/tcp closed   ms-wbt-server
MAC Address: DE:AD:00:00:BE:EF (Intel Corporate)

Nmap done: 1 IP address (1 host up) scanned in 1.44 seconds
```

Scanning Options	Description
`10.129.2.28`	Scans the specified target.
`--top-ports=10`	Scans the specified top ports that have been defined as most frequent.

We see that we only scanned the top 10 TCP ports of our target, and `Nmap` displays their state accordingly. If we trace the packets `Nmap` sends,

we will see the `RST` flag on `TCP port 21` that our target sends back to us. To have a clear view of the SYN scan, we disable the ICMP echo requests (`-Pn`), DNS resolution (`-n`), and ARP ping scan (`--disable-arp-ping`).

Nmap - Trace the Packets

```
pecna@htb[/htb]$ sudo nmap 10.129.2.28 -p 21 --packet-trace
-Pn -n --disable-arp-ping

Starting Nmap 7.80 ( https://nmap.org ) at 2020-06-15 15:39
CEST
SENT (0.0429s) TCP 10.10.14.2:63090 > 10.129.2.28:21 S ttl=
56 id=57322 iplen=44  seq=1699105818 win=1024 <mss 1460>
RCVD (0.0573s) TCP 10.129.2.28:21 > 10.10.14.2:63090 RA ttl
=64 id=0 iplen=40  seq=0 win=0
Nmap scan report for 10.129.2.28
Host is up (0.014s latency).

PORT   STATE  SERVICE
21/tcp closed ftp
MAC Address: DE:AD:00:00:BE:EF (Intel Corporate)

Nmap done: 1 IP address (1 host up) scanned in 0.07 seconds
```

Scanning Options	Description
`10.129.2.28`	Scans the specified target.
`-p 21`	Scans only the specified port.
`--packet-trace`	Shows all packets sent and received.
`-n`	Disables DNS resolution.
`--disable-arp-ping`	Disables ARP ping.

We can see from the SENT line that we (`10.10.14.2`) sent a TCP packet with the `SYN` flag (`S`) to our target (`10.129.2.28`). In the next RCVD line, we can see that the target responds with a TCP packet containing the `RST` and `ACK` flags (`RA`). `RST` and `ACK` flags are used to acknowledge receipt of the TCP packet (`ACK`) and to end the TCP session (`RST`).

Request

Message	Description
`SENT (0.0429s)`	Indicates the SENT operation of Nmap, which sends a packet to the target.
`TCP`	Shows the protocol that is being used to interact with the target port.
`10.10.14.2:63090 >`	Represents our IPv4 address and the source port, which will be used by Nmap to send the packets.
`10.129.2.28:21`	Shows the target IPv4 address and the target port.
`S`	SYN flag of the sent TCP packet.
`ttl=56 id=57322 iplen=44 seq=1699105818 win=1024 mss 1460`	Additional TCP Header parameters.

Response

Message	Description
`RCVD (0.0573s)`	Indicates a received packet from the target.
`TCP`	Shows the protocol that is being used.
`10.129.2.28:21 >`	Represents targets IPv4 address and the source port, which will be used to reply.
`10.10.14.2:63090`	Shows our IPv4 address and the port that will be replied to.
`RA`	RST and ACK flags of the sent TCP packet.
`ttl=64 id=0 iplen=40 seq=0 win=0`	Additional TCP Header parameters.

Connect Scan

The Nmap TCP Connect scan (`-sT`) performs a full three-way TCP handshake (SYN → SYN-ACK → ACK) to determine port state—if the target replies SYN-ACK the port is **open**, if it replies RST the port is **closed**—making it highly accurate. Because it completes real connections it is one of the **least stealthy** scan types: it leaves normal connection logs and is readily detected by IDS/IPS, though it's relatively "polite" to services and unlikely to crash them since it behaves like a normal client. It's slower than lighter scans because each connection must be completed and responses waited for, but it's useful when accuracy matters or when a host's firewall blocks unsolicited incoming packets yet allows outgoing connections. For greater stealth (but slightly less certainty)

people commonly use SYN (half-open) scans, which do not finish the handshake.

Connect Scan on TCP Port 443

```
pecna@htb[/htb]$ sudo nmap 10.129.2.28 -p 443 --packet-trac
e --disable-arp-ping -Pn -n --reason -sT

Starting Nmap 7.80 ( https://nmap.org ) at 2020-06-15 16:26
CET
CONN (0.0385s) TCP localhost > 10.129.2.28:443 => Operation
now in progress
CONN (0.0396s) TCP localhost > 10.129.2.28:443 => Connected
Nmap scan report for 10.129.2.28
Host is up, received user-set (0.013s latency).

PORT    STATE SERVICE REASON
443/tcp open  https   syn-ack

Nmap done: 1 IP address (1 host up) scanned in 0.04 seconds
```

Filtered Ports

When a port is shown as filtered, it can have several reasons. In most cases, firewalls have certain rules set to handle specific connections. The packets can either be `dropped`, or `rejected`. When a packet gets dropped, `Nmap` receives no response from our target, and by default, the retry rate (`--max-retries`) is set to `10`. This means `Nmap` will resend the request to the target port to determine if the previous packet was accidentally mishandled or not.

Let us look at an example where the firewall `drops` the TCP packets we send for the port scan. Therefore we scan the TCP port **139**, which was already shown as filtered. To be able to track how our sent packets are handled, we deactivate the ICMP echo requests (`-Pn`), DNS resolution (`-n`), and ARP ping scan (`--disable-arp-ping`) again.

Host and Port Scanning

```
pecna@htb[/htb]$ sudo nmap 10.129.2.28 -p 139 --packet-trac
e -n --disable-arp-ping -Pn

Starting Nmap 7.80 ( https://nmap.org ) at 2020-06-15 15:45
CEST
SENT (0.0381s) TCP 10.10.14.2:60277 > 10.129.2.28:139 S ttl
=47 id=14523 iplen=44  seq=4175236769 win=1024 <mss 1460>
SENT (1.0411s) TCP 10.10.14.2:60278 > 10.129.2.28:139 S ttl
=45 id=7372 iplen=44  seq=4175171232 win=1024 <mss 1460>
Nmap scan report for 10.129.2.28
Host is up.

PORT    STATE    SERVICE
139/tcp filtered netbios-ssn
MAC Address: DE:AD:00:00:BE:EF (Intel Corporate)

Nmap done: 1 IP address (1 host up) scanned in 2.06 seconds
```

Scanning Options	Description
`10.129.2.28`	Scans the specified target.
`-p 139`	Scans only the specified port.
`--packet-trace`	Shows all packets sent and received.
`-n`	Disables DNS resolution.
`--disable-arp-ping`	Disables ARP ping.
`-Pn`	Disables ICMP Echo requests.

We see in the last scan that `Nmap` sent two TCP packets with the SYN flag. By the duration (`2.06s`) of the scan, we can recognize that it took much longer than the previous ones (`~0.05s`). The case is different if the firewall rejects the packets. For this, we look at TCP port `445`, which is handled accordingly by such a rule of the firewall.

```
pecna@htb[/htb]$ sudo nmap 10.129.2.28 -p 445 --packet-trac
e -n --disable-arp-ping -Pn

Starting Nmap 7.80 ( https://nmap.org ) at 2020-06-15 15:55
CEST
```

```
SENT (0.0388s) TCP 10.129.2.28:52472 > 10.129.2.28:445 S tt
l=49 id=21763 iplen=44  seq=1418633433 win=1024 <mss 1460>
RCVD (0.0487s) ICMP [10.129.2.28 > 10.129.2.28 Port 445 unr
eachable (type=3/code=3) ] IP [ttl=64 id=20998 iplen=72 ]
Nmap scan report for 10.129.2.28
Host is up (0.0099s latency).

PORT    STATE    SERVICE
445/tcp filtered microsoft-ds
MAC Address: DE:AD:00:00:BE:EF (Intel Corporate)

Nmap done: 1 IP address (1 host up) scanned in 0.05 seconds
```

Scanning Options	Description
`10.129.2.28`	Scans the specified target.
`-p 445`	Scans only the specified port.
`--packet-trace`	Shows all packets sent and received.
`-n`	Disables DNS resolution.
`--disable-arp-ping`	Disables ARP ping.
`-Pn`	Disables ICMP Echo requests.

As a response, we receive an `ICMP` reply with `type 3` and `error code 3`, which indicates that the desired port is unreachable. Nevertheless, if we know that the host is alive, we can strongly assume that the firewall on this port is rejecting the packets, and we will have to take a closer look at this port later.

Discovering Open UDP Ports

Some system administrators sometimes forget to filter the UDP ports in addition to the TCP ones. Since UDP is a stateless protocol and does not require a three-way handshake like TCP. We do not receive any acknowledgment. Consequently, the timeout is much longer, making the whole UDP scan (-sU) much slower than the TCP scan (-sS).

Let's look at an example of what a UDP scan (`-sU`) can look like and what results it gives us.

```
pecna@htb[/htb]$ sudo nmap 10.129.2.28 -F -sUStarting Nmap
7.80 ( https://nmap.org ) at 2020-06-15 16:01 CEST
Nmap scan report for 10.129.2.28
Host is up (0.059s latency).
Not shown: 95 closed ports
PORT     STATE         SERVICE
68/udp   open|filtered dhcpc
137/udp  open          netbios-ns
138/udp  open|filtered netbios-dgm
631/udp  open|filtered ipp
5353/udp open          zeroconf
MAC Address: DE:AD:00:00:BE:EF (Intel Corporate)

Nmap done: 1 IP address (1 host up) scanned in 98.07 second
s
```

Scanning Options	Description
`10.129.2.28`	Scans the specified target.
`-F`	Scans top 100 ports.
`-sU`	Performs a UDP scan.

Another disadvantage of this is that we often do not get a response back because `Nmap` sends empty datagrams to the scanned UDP ports, and we do not receive any response. So we cannot determine if the UDP packet has arrived at all or not. If the UDP port is `open`, we only get a response if the application is configured to do so.

```
pecna@htb[/htb]$ sudo nmap 10.129.2.28 -sU -Pn -n --disable
-arp-ping --packet-trace -p 137 --reason

Starting Nmap 7.80 ( https://nmap.org ) at 2020-06-15 16:15
CEST
SENT (0.0367s) UDP 10.10.14.2:55478 > 10.129.2.28:137 ttl=5
7 id=9122 iplen=78
RCVD (0.0398s) UDP 10.129.2.28:137 > 10.10.14.2:55478 ttl=6
4 id=13222 iplen=257
Nmap scan report for 10.129.2.28
```

```
Host is up, received user-set (0.0031s latency).

PORT    STATE SERVICE    REASON
137/udp open  netbios-ns udp-response ttl 64
MAC Address: DE:AD:00:00:BE:EF (Intel Corporate)

Nmap done: 1 IP address (1 host up) scanned in 0.04 seconds
```

Scanning Options	Description
`10.129.2.28`	Scans the specified target.
`-sU`	Performs a UDP scan.
`-Pn`	Disables ICMP Echo requests.
`-n`	Disables DNS resolution.
`--disable-arp-ping`	Disables ARP ping.
`--packet-trace`	Shows all packets sent and received.
`-p 137`	Scans only the specified port.
`--reason`	Displays the reason a port is in a particular state.

If we get an ICMP response with `error code 3` (port unreachable), we know that the port is indeed `closed`.

```
pecna@htb[/htb]$ sudo nmap 10.129.2.28 -sU -Pn -n --disable
-arp-ping --packet-trace -p 100 --reason

Starting Nmap 7.80 ( https://nmap.org ) at 2020-06-15 16:25
CEST
SENT (0.0445s) UDP 10.10.14.2:63825 > 10.129.2.28:100 ttl=5
7 id=29925 iplen=28
RCVD (0.1498s) ICMP [10.129.2.28 > 10.10.14.2 Port unreacha
ble (type=3/code=3) ] IP [ttl=64 id=11903 iplen=56 ]
Nmap scan report for 10.129.2.28
Host is up, received user-set (0.11s latency).

PORT    STATE  SERVICE REASON
100/udp closed unknown port-unreach ttl 64
MAC Address: DE:AD:00:00:BE:EF (Intel Corporate)
```

```
Nmap done: 1 IP address (1 host up) scanned in  0.15 second
s
```

Scanning Options	Description
`10.129.2.28`	Scans the specified target.
`-sU`	Performs a UDP scan.
`-Pn`	Disables ICMP Echo requests.
`-n`	Disables DNS resolution.
`--disable-arp-ping`	Disables ARP ping.
`--packet-trace`	Shows all packets sent and received.
`-p 100`	Scans only the specified port.
`--reason`	Displays the reason a port is in a particular state.

For all other ICMP responses, the scanned ports are marked as (`open|filtered`).

```
pecna@htb[/htb]$ sudo nmap 10.129.2.28 -sU -Pn -n --disable
-arp-ping --packet-trace -p 138 --reason

Starting Nmap 7.80 ( https://nmap.org ) at 2020-06-15 16:32
CEST
SENT (0.0380s) UDP 10.10.14.2:52341 > 10.129.2.28:138 ttl=5
0 id=65159 iplen=28
SENT (1.0392s) UDP 10.10.14.2:52342 > 10.129.2.28:138 ttl=4
0 id=24444 iplen=28
Nmap scan report for 10.129.2.28
Host is up, received user-set.

PORT    STATE         SERVICE     REASON
138/udp open|filtered netbios-dgm no-response
MAC Address: DE:AD:00:00:BE:EF (Intel Corporate)

Nmap done: 1 IP address (1 host up) scanned in 2.06 seconds
```

Scanning Options	Description
`10.129.2.28`	Scans the specified target.
`-sU`	Performs a UDP scan.

Scanning Options	Description
`-Pn`	Disables ICMP Echo requests.
`-n`	Disables DNS resolution.
`--disable-arp-ping`	Disables ARP ping.
`--packet-trace`	Shows all packets sent and received.
`-p 138`	Scans only the specified port.
`--reason`	Displays the reason a port is in a particular state.

Another handy method for scanning ports is the `-sV` option which is used to get additional available information from the open ports. This method can identify versions, service names, and details about our target.

```
pecna@htb[/htb]$ sudo nmap 10.129.2.28 -Pn -n --disable-arp
-ping --packet-trace -p 445 --reason  -sV

Starting Nmap 7.80 ( https://nmap.org ) at 2022-11-04 11:10
GMT
SENT (0.3426s) TCP 10.10.14.2:44641 > 10.129.2.28:445 S ttl
=55 id=43401 iplen=44  seq=3589068008 win=1024 <mss 1460>
RCVD (0.3556s) TCP 10.129.2.28:445 > 10.10.14.2:44641 SA tt
l=63 id=0 iplen=44  seq=2881527699 win=29200 <mss 1337>
NSOCK INFO [0.4980s] nsock_iod_new2(): nsock_iod_new (IOD#
1)NSOCK INFO [0.4980s] nsock_connect_tcp(): TCP connection
requested to 10.129.2.28:445 (IOD#1) EID 8NSOCK INFO [0.513
0s] nsock_trace_handler_callback(): Callback: CONNECT SUCCE
SS for EID 8 [10.129.2.28:445]
Service scan sending probe NULL to 10.129.2.28:445 (tcp)
NSOCK INFO [0.5130s] nsock_read(): Read request from IOD#1
[10.129.2.28:445] (timeout: 6000ms) EID 18NSOCK INFO [6.519
0s] nsock_trace_handler_callback(): Callback: READ TIMEOUT
for EID 18 [10.129.2.28:445]
Service scan sending probe SMBProgNeg to 10.129.2.28:445 (t
cp)
NSOCK INFO [6.5190s] nsock_write(): Write request for 168 b
ytes to IOD#1 EID 27 [10.129.2.28:445]NSOCK INFO [6.5190s]
nsock_read(): Read request from IOD#1 [10.129.2.28:445] (ti
meout: 5000ms) EID 34NSOCK INFO [6.5190s] nsock_trace_handl
er_callback(): Callback: WRITE SUCCESS for EID 27 [10.129.
```

```
2.28:445]
NSOCK INFO [6.5320s] nsock_trace_handler_callback(): Callba
ck: READ SUCCESS for EID 34 [10.129.2.28:445] (135 bytes)
Service scan match (Probe SMBProgNeg matched with SMBProgNe
g line 13836): 10.129.2.28:445 is netbios-ssn.  Version: |S
amba smbd|3.X - 4.X|workgroup: WORKGROUP|
NSOCK INFO [6.5320s] nsock_iod_delete(): nsock_iod_delete
(IOD#1)Nmap scan report for 10.129.2.28
Host is up, received user-set (0.013s latency).

PORT    STATE SERVICE     REASON         VERSION
445/tcp open  netbios-ssn syn-ack ttl 63 Samba smbd 3.X -
4.X (workgroup: WORKGROUP)
Service Info: Host: Ubuntu

Service detection performed. Please report any incorrect re
sults at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 6.55 seconds
```

Scanning Options	Description
`10.129.2.28`	Scans the specified target.
`-Pn`	Disables ICMP Echo requests.
`-n`	Disables DNS resolution.
`--disable-arp-ping`	Disables ARP ping.
`--packet-trace`	Shows all packets sent and received.
`-p 445`	Scans only the specified port.
`--reason`	Displays the reason a port is in a particular state.
`-sV`	Performs a service scan.

Network_Enumeration_With_Nmap_Module_Cheat_Sheet (1).pdf